

L2 Regularization (Weight Decay)

Deep Learning

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Contents of This Video

In this video, we will cover:

- The idea behind L2 regularization
- Mathematical formulation
- The weight decay update rule
- Effect on model complexity
- Connection to ridge regression

The Core Idea

Observation: Overfit models often have large weights

Solution: Penalize large weights in the loss function

$$L_{reg} = L_{data} + \lambda \sum_i w_i^2$$

Where:

- L_{data} is the original loss (MSE, cross-entropy, etc.)
- λ is the regularization strength
- $\sum_i w_i^2$ is the sum of squared weights

Mathematical Formulation

Original loss:

$$L = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$$

Regularized loss:

$$L_{reg} = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i) + \lambda \|W\|_2^2$$

Where $\|W\|_2^2 = \sum_l \sum_{i,j} (W_{ij}^{(l)})^2$ sums over all weights

The Gradient with L2 Regularization

Original gradient:

$$\frac{\partial L}{\partial w} = \frac{\partial L_{data}}{\partial w}$$

Regularized gradient:

$$\frac{\partial L_{reg}}{\partial w} = \frac{\partial L_{data}}{\partial w} + 2\lambda w$$

The gradient includes the current weight value!

Weight Decay Update Rule

Standard update:

$$w \leftarrow w - \alpha \frac{\partial L_{data}}{\partial w}$$

With L2 regularization:

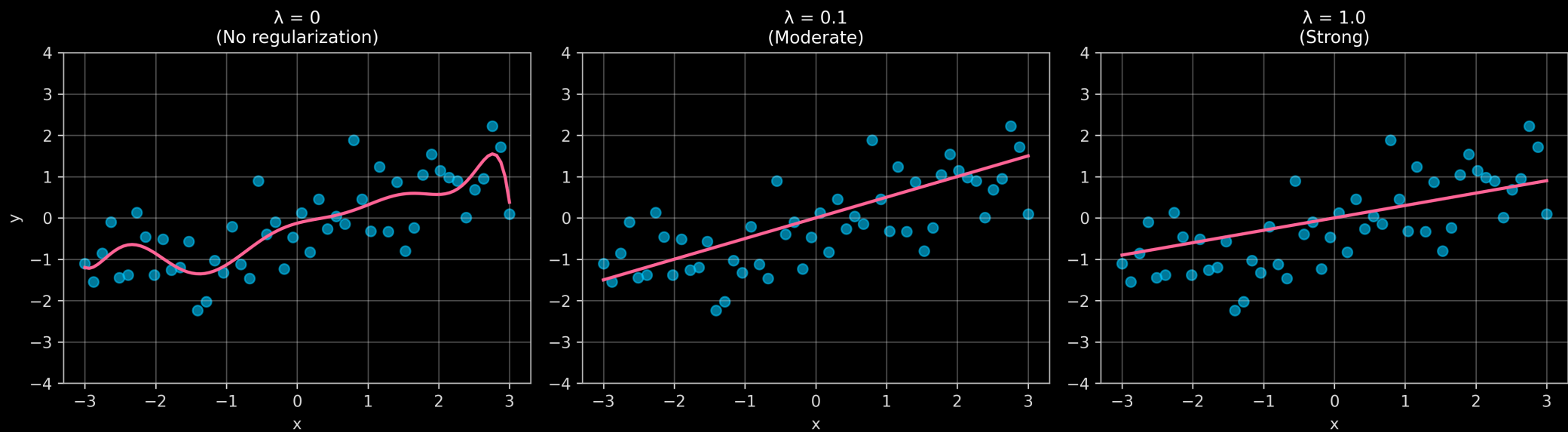
$$w \leftarrow w - \alpha \left(\frac{\partial L_{data}}{\partial w} + 2\lambda w \right)$$

Rearranging:

$$w \leftarrow (1 - 2\alpha\lambda)w - \alpha \frac{\partial L_{data}}{\partial w}$$

The weight is multiplied by $(1 - 2\alpha\lambda) < 1$ each step: it decays!

Effect of λ



Connection to Ridge Regression

Ridge regression = Linear regression + L2 regularization

$$L = \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \lambda \|\mathbf{w}\|_2^2$$

Same principle applies to neural networks!

- Ridge shrinks coefficients toward zero
- L2 reg in NNs shrinks weights toward zero
- Both reduce model complexity

Practical Considerations

Typical values: $\lambda \in [10^{-5}, 10^{-2}]$

Tips:

- Start with $\lambda = 0.01$, adjust based on validation loss
- Higher λ = more regularization = simpler model
- Monitor train/val gap: if large, increase λ
- Can use different λ for different layers

What We've Covered

In this video, we've learned:

- L2 regularization adds $\lambda \sum w_i^2$ to the loss
- Gradient becomes: $\nabla L_{data} + 2\lambda w$
- Update rule shrinks weights: $w \leftarrow (1 - 2\alpha\lambda)w - \alpha\nabla L$
- Larger λ = more regularization = simpler model
- Same principle as ridge regression